



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

ENG 111: English Basics

Mid-term Examination
Spring 2026

30th Batch

17 June 2026

1st Semester

Time: 01 Hour and 30 Minutes

Total Marks: 30

CLOs

Marks

PART A

CLO5 01. Complete each sentence below by filling in the blank with the most appropriate verb form that agrees with the subject.

0.5x10 =05

- a) The jury is (be) arguing over the case.
- b) The outcome, not the planning, is (be) important.
- c) The police have (have) investigated the case seriously.
- d) Neither my brother nor my parents are (be) willing to help me.
- e) It is I who am (have) faced everything alone.
- f) Pride and Prejudice is (worth) being masterpiece.
- g) The committee (meet) every Saturday.
- h) The great singer, along with his team, is (be) invited to the show.
- i) The government but not the citizens are (be) responsible for this case.
- j) Twenty minutes is (be) required to complete the task.

CLO5 02. Transform the given sentences into the specified forms as indicated.

1x5 =05

- a) Has the work been finished by you? (Active)
- b) Walk fast or you will miss the bus. (Simple)
- c) Due to illness, he stayed home. (Complex)
- d) Do not waste time. (Passive)
- e) Being honest, he is respected. (Compound)

walking
fast,

PART B

Read the passage below and answer the following questions:

Social media platforms have become an essential part of modern communication. Every day, millions of users share photos, videos, messages, and opinions online. To manage this enormous amount of information efficiently, [social media companies rely on data structures and algorithms. Data structures

ill,

are methods of organizing and storing data so that it can be accessed quickly. User profiles, friend lists, comments, and messages are stored in ways that allow platforms to retrieve information efficiently. Without proper organization, finding specific content among billions of posts would be extremely difficult. Algorithms are sets of instructions that help computers analyze data and make decisions. On social media platforms, algorithms determine which posts appear in a user's feed, suggest new friends, recommend videos, and select content that is most likely to attract attention. They analyze user behavior, including likes, comments, shares, and viewing time, to create a personalized experience for each user. While these technologies improve efficiency and user engagement, they can also encourage excessive use of social media. Algorithms are often designed to keep users active on the platform by continuously recommending interesting content. As users receive likes, notifications, and entertaining videos, the brain releases dopamine, a chemical associated with pleasure and reward. This creates a desire to keep scrolling and checking for new content. As a result, many people spend more time on social media than they intend. Frequent exposure to short and rapidly changing content may also reduce attention span, making it harder to focus on long reading tasks, academic work, or other activities that require sustained concentration. Therefore, although data structures and algorithms make social media efficient and engaging, users should be aware of their effects and use these platforms responsibly.

CLO2 03. Choose the most suitable option for each of the following questions.

1x5 =05

a) Why would finding specific content be difficult without data structures?

- i. Because social media platforms would contain fewer users.
- ☒ ii. Because information would not be organized for efficient retrieval.
- iii. Because algorithms would stop recommending content.
- iv. Because users would not be able to upload posts.

b) What can be inferred about social media algorithms from the passage?

- i. Their primary function is to reduce user activity.
- ☒ ii. They select content based partly on users' previous interactions.
- iii. They only recommend content created by friends.
- iv. They are designed mainly to store user profiles.

c) Why is dopamine mentioned in the passage?

- i. To explain how social media platforms earn money.
- ii. To describe a chemical used in computer programming.
- ☒ iii. To show why users may repeatedly return to social media.
- iv. To prove that all social media use is harmful.

d) Which statement best reflects the relationship between algorithms and excessive social media use?

- i. Algorithms discourage users from viewing additional content.
- ii. Algorithms have no influence on how long users remain online.
- ☒ iii. Algorithms may increase user engagement by appealing content.

iv. Algorithms force users to stay connected against their will.

e) Which of the following is most strongly suggested by the passage?

i. Personalized content may contribute to reduced concentration. ~~X~~

☒ ii. Data structures are responsible for the release of dopamine.

iii. Social media companies are unable to analyze user behavior.

iv. Short videos improve users' ability to focus for long periods. ~~X~~

CLO2 04. Fill in the blanks with appropriate words.

1x5=05

- a) Data structures help social media platforms store and access information more
- b) Algorithms analyze user to create a personalized experience.
- c) Notifications, likes, and entertaining videos can trigger the release of in the brain.
- d) Many users spend more time on social media than they to.
- e) Frequent exposure to short and rapidly changing content may reduce a person's span.

CLO2 05. Answer the following questions providing information from the passage. 2x5=10

- a) Why do social media platforms rely on data structures?
- ☒ b) How do algorithms create a personalized experience for users?
- ☒ c) Why do users often continue scrolling through social media content?
- ☒ d) According to the passage, how can excessive social media use affect attention span?
- e) What does the passage suggest users should do to avoid the negative effects of social media?



Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. The symbols used have their usual meaning.]

		CLOs	Marks
Q1.	(a) What is the definition of free vibration?	CLO1	[1]
	(b) Give some characteristics of SHM.	CLO1	[4]
	(c) Derive the differential equation for a particle executing simple harmonic motion.	CLO2	[3]
	(d) For a particle executing simple harmonic motion the displacement is 8 cm at the instant the velocity is 6 cm/sec and the displacement is 6 cm at the instant the velocity is 8 cm/sec. Calculate (i) amplitude, (ii) frequency and (iii) time period.	CLO3	[2]
			$A = 8$ $V = 6$
Q2.	(a) Define the following terms i. SHM ii. Time period. iii. Angular frequency. iv. Two-body oscillation.	CLO1	[4]
	(b) Solve the differential equation to obtain an expression for the displacement of SHM.	CLO2	[4]
	(c) Show that the time period of a particle executing SHM is equal to 2π times the square root of the ratio of displacement to acceleration.	CLO2	[2]
Q3.	(a) Define forced oscillation and concisely explain how it works.	CLO2	[2]
	(b) Obtain the mathematical equations for velocity and acceleration in a system undergoing simple harmonic motion.	CLO2	[4]
	(c) Derive the differential equation for a damped harmonic oscillator.	CLO2	[4]
Q4.	(a) What is resonance?	CLO1	[1]
	(b) Derive the expression for the reduced mass of a two-body oscillator.	CLO2	[4]
	(c) The scale of a spring balance reading from 0 to 10kg is 0.25 m long. A body suspended from the balance is found to oscillate vertically with a frequency of $10/\pi$ Hz. Calculate the mass of the body attached to the spring.	CLO3	[3]
	(d) What is meant by the term logarithmic decrement of a damped harmonic oscillator? Obtain an expression for this term.	CLO1 +	[2]
		CLO2	
Q5.	(a) Explain the laws of reflection and refraction with suitable diagrams.	CLO2	[3]
	(b) Derive the Smith-Helmholtz equation and deduce Lagrange's law from it.	CLO2	[7]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 30th

Date: 10/06/2026

Mid-Term Examination

Semester: Spring 2026

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

5 x 6 = 30

1. (a) Write short notes on Bi-Conditional and Contingency of a Proposition. CLO 1 2
(b) By using proposition, proof De-Morgan's law for three bits. CLO 1 4
2. (a) State different types of function with diagram. CLO 4 3
(b) Let $f_1(x) = x^3 - 2x^2 + 1$ and $f_2(x) = x^3 + 4x - 2$. Determine the types of given functions with reason. Find $f_1 f_2$ and $f_1 + f_2$. CLO 4 3
3. (a) Determine the general equation for the Pizza Cut Problem. CLO 1 2
(b) By using Mathematical Induction technique, prove that $1^2 + 2^2 + 3^2 + \dots + n^2 = (n(n+1)(2n+1))/6$. CLO 1 4
4. (a) Define Universal quantifier and Existential quantifier. CLO 1 2
(b) What is meant by Recurrence relation? Find the first five terms of the sequence defined by the recurrence relation and initial condition given below: $a_n = na_{n-1} + n^2 a_{n-2}$, $a_0 = 2$, $a_1 = 2$. CLO 4 4
5. Determine the expressions $\{[p \wedge \neg(\neg p \vee (q \oplus \neg r)) \leftrightarrow (p \wedge q)] \rightarrow p\}$ and $\{[\neg(p \vee (p \oplus \neg q)) \leftrightarrow (q \wedge \neg r)]\}$ are either logically equivalent or not. CLO 1 6
6. Find the truth values for the following expression: $[(\neg p \vee q) \oplus (\neg p \vee r)] \leftrightarrow ((q \wedge r) \rightarrow (p \leftrightarrow \neg r)) \rightarrow \neg q$, and justify whether the given expression is either a Tautology or not. CLO 1 6
7. Let $A = \{1, 2, 3, 6, 12, 24\}$ be a set, find the following relations: CLO 4 6
 $R_1 = \{(a, b) \mid a \leq b\}$,
 $R_2 = \{(a, b) \mid a \text{ divides } b\}$ and
 $R_3 = \{(a, b) \mid a^2 + b \leq 50\}$
Find the properties with reason of each relation. Represent each relation in matrix form and graphically.



STANDARD UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Faculty of Science, Engineering and Technology

Course Code: MAT115

Course Title: Differential and Integral Calculus

Mid-Term Examination

Semester: Spring 2026

Date: 6/20/2026

Total Time: 1:30 Hours

Total Marks: 30

Answer any Three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1.	(a) Define differentiability of a function at a point.	CLO1	[1]
	(b) Prove that every differentiable function is always continuous.	CLO2	[2]
	(c) A function $f(x)$ is defined as:	CLO3	[7]
	$f(x) = \begin{cases} 1 + 2x; & -\frac{1}{2} \leq x < 0 \\ 1 - 2x; & 0 \leq x \leq \frac{1}{2} \\ -1 + 2x; & x > \frac{1}{2} \end{cases}$		
	(i) Determine the domain and range of $f(x)$.		
	(ii) Test the continuity and differentiability of $f(x)$ at $x = \frac{1}{2}$.		
Q2.	(a) State and prove Leibnitz theorem.	CLO2	[4]
	(b) Determine $\frac{dy}{dx}$ for the following (any two):	CLO3	[6]
	(i) $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$		
	(ii) $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$		
	(iii) $e^{xy} - 4xy = 1$		
Q3.	(a) What is an inflection point?	CLO1	[1]
	(b) If $y = \cos(m \sin^{-1} x)$, then prove that	CLO3	[5]
	(i) $(1 - x^2)y_2 - xy_1 + m^2y = 0$		
	(ii) $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - m^2)y_n = 0$		
	(c) Show that the function $f(x) = x^5 - 6x^2 + 12x - 3$ has neither maximum nor a minimum when $x = 2$.	CLO3	[4]
Q4.	(a) Define homogeneous function of three variables.	CLO1	[1]
	(b) State and prove the Euler theorem for homogeneous function of two variables.	CLO2	[5]
	(c) Verify Euler theorem for the function	CLO3	[4]
	$f(x, y, z) = x^5 + y^5 + z^5 - x^3yz - xy^3z - xyz^3$.		
Q5.	(a) Define partial differentiation of a function of two variables.	CLO1	[1]
	(b) If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$, then show that	CLO3	[4]
	(i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$		
	(ii) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3$		
	(c) If $u(x, y, z) = \ln(x^2 + y^2 + z^2)$, then prove that $xu_{yz} = yu_{zx} = zu_{xy}$.	CLO3	[5]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 117: Engineering Ethics and Cyber Law

Batch: 30th

Semester: Spring 2026

Date: 13/06/2026

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

- ① (a) What is Moral Dilemma? CLO1 3
(b) Outline the differences between Morality and Ethics. CLO1 3
- ② Alex, a system engineer, discovers that his company's new IoT device collects more personal user data than disclosed in their privacy policy. The data is being sold to third parties without user consent. When he raises this concern, his managers argue that all tech companies do this and it's essential for their business model.
(a) Evaluate this situation using Ethical Theories. CLO2 3
(b) Under what conditions would whistleblowing be ethically justified in this case? Support your answer with ethical principles CLO2 3
- ③ Think about a scenario, you are giving exam on this course and you didn't study the chapters properly. So naturally you are biting your pen instead of writing by it on the paper. Now suddenly you see your that same friend who told you last night that "Mama, kichuuuu Pori nai. Kal Sure fail" is writing smoothly at the speed of light!! You also notice that the secret of his success is not his study rather his ability to do copy and bring 'Nokol' like a pro in the exam hall. CLO2 6

Now explain this real-life part of yours from the 3 normative ethical approaches that we discussed on the class.

4. (a) Explain Aristotle's Golden Mean using an example of Balance Personality CLO2 3
(b) How does MacIntyre define virtues in the context of professional practices? CLO2 3
- ⑤ Explain the Lawrence Kohlberg's Theory on moral development. CLO2 6
- ⑥ An engineer discovers a small but critical error in a project. Reporting it may delay the launch and harm the company's image.
(a) What kind of ethical dilemma is this? Explain your answer. CLO2 3
(b) Analyze the situation using duty ethics and consequentialism. CLO2 3
7. (a) Outline the differences between morality and ethics. CLO1 3
(b) Define utilitarianism and give a real-life example. CLO1 3



HAMDARD UNIVERSITY BANGLADESH

Department of Electrical and Electronics Engineering

Faculty of Science, Engineering and Technology

Course Code: EEE117

Course Title: Electrical Engineering

Mid-Term Examination

Semester: Spring 2026

Date: 6/22/2026

Total Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any three (03) of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1 (a) Define the following questions:

2 CLO1

- I. Charge
- II. Current
- III. Voltage
- IV. Power

(b) Calculate the power supplied or absorbed by each element in figure-1(b).

4 CLO3

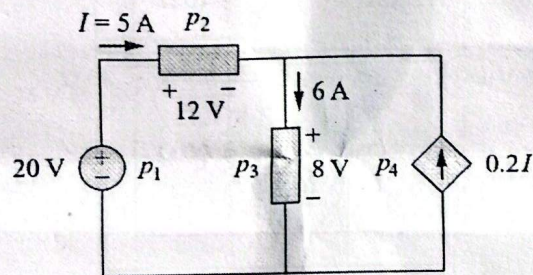


Figure-1(b)

$$P = VI$$

$$P = IV$$

0.2

(c) Calculate the equivalent resistance in the circuit in Figure-1(c):

4 CLO3

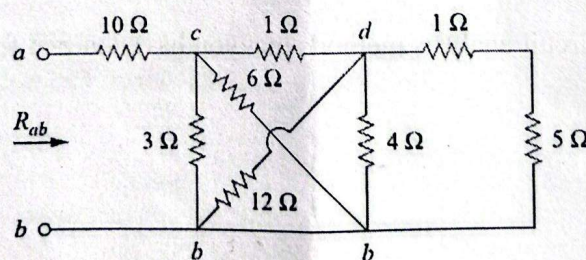


Figure-1(c)

2 (a) State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

2 CLO2

(b) Find currents and voltages in the circuit shown in Figure-2(b):

4 CLO3

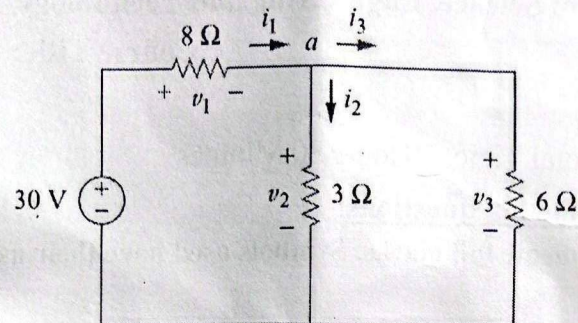


Figure-2(b)

(c) Obtain the equivalent resistance R_{ab} for the circuit in Figure-2(c) and use it to find current i .

$$i = \frac{V}{R}$$

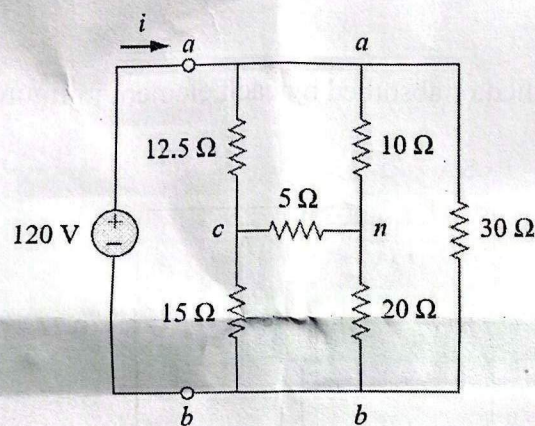


Figure-2(c)

3 (a) Demonstrate the circuit analysis methods that you have learned from your class. 2 CLO2

(b) Find v_1 , v_2 and v_3 in the circuit of Figure-3(b) using nodal analysis.

4 CLO3

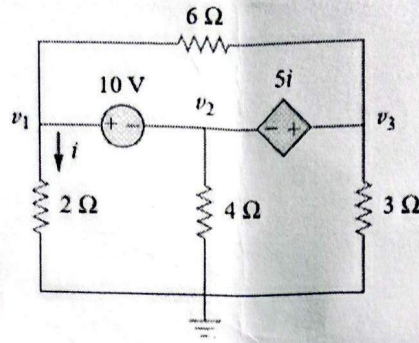


Figure-3(b)

$$\frac{v_1 - v_0}{2} = \frac{5}{2}$$

(c) Use mesh analysis to find the current I_0 in the circuit of Figure-3(c).

4 CLO3

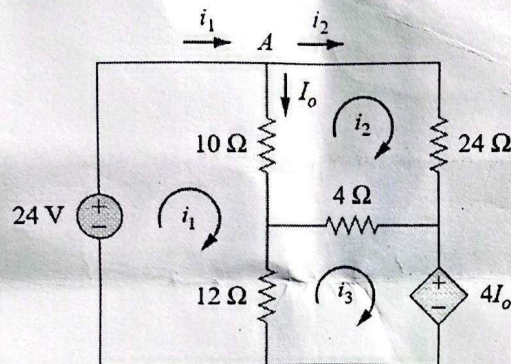


Figure-3(c)

4 (a) Write a short note on following questions.

2 CLO1

- I. Thevenin theorem
- II. Superposition Theorem

(b) (i) Find i_0 in the circuit of Figure-4(b-i) using source transformation.

8 CLO3

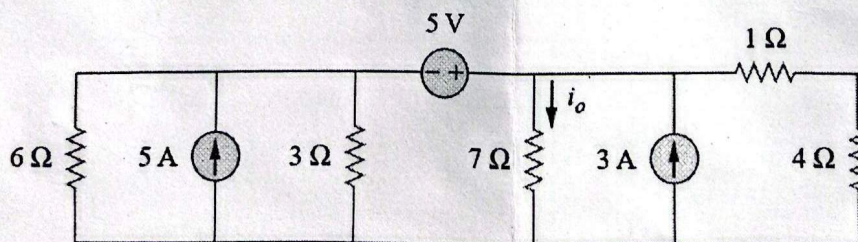


Figure-4(b-i)

$$\frac{2 \times 2.4 \times 3}{1, 2, 3}$$

(ii) Find the Thevenin equivalent of the circuit in Figure-4(b-ii) at terminals a-b.

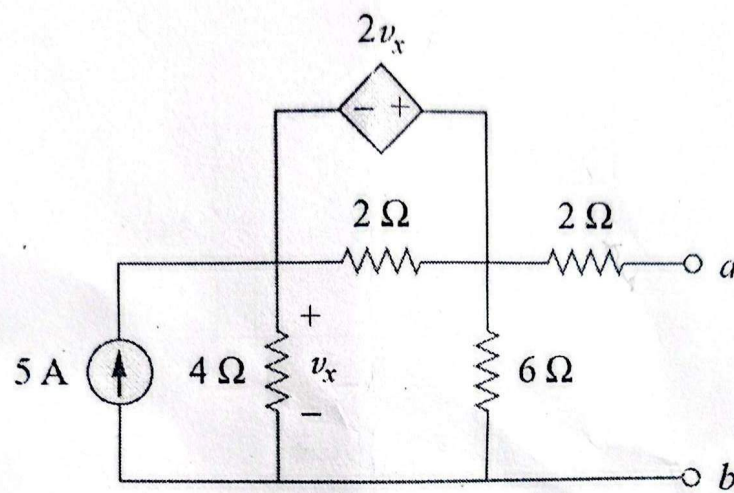


Figure-4(b-ii)